Indefinite Integral

Lecture 2

Recap

We already know the antiderivative of a function f. The process of finding antiderivatives is called **antidifferentiation** or **integration**.

Definition

If $F(x)$ is the antiderivative of $f(x)$, then

$$\int f(x)\,dx = F(x) + C,$$

where C is the integral constant.

Example

Evaluate the following indefinite integral: $\int (x^4 + 3x - 9)\,dx$

Ans: $\int (x^4 + 3x - 9)\,dx = \dfrac{x^5}{5} + \dfrac{3}{2}x^2 - 9x + C$

Integration Formulas

Book: *Calculus: Early Transcendentals* — H. Anton, page 324, Table 5.2.1.

Some standard formulas

$$\int dx = x + C$$

$$\int x^r\,dx = \frac{x^{r+1}}{r+1} + C, \quad r \neq -1$$

$$\int \cos x\,dx = \sin x + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \sec^2 x\,dx = \tan x + C$$

$$\int \csc^2 x\,dx = -\cot x + C$$

and so on.

Properties

Suppose that $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$ respectively, and a is a constant. Then:

Properties

(a) $\int af(x)\,dx = aF(x) + C$

(b) $\int [f(x) \pm g(x)]\,dx = F(x) \pm G(x) + C$

Example

$$\begin{aligned}\int (3x^6 - 2x^2 + 7x)\,dx &= 3\int x^6\,dx - 2\int x^2\,dx + 7\int x\,dx \\ &= 3\frac{x^{6+1}}{6+1} - 2\frac{x^{2+1}}{2+1} + 7\frac{x^2}{2} + C \\ &= \frac{3}{7}x^7 - \frac{2}{3}x^3 + \frac{7}{2}x^2 + C\end{aligned}$$

Problem: Evaluate

1. $\displaystyle\int \frac{\cos x}{\sin^2 x}\,dx$

$$\begin{aligned}\int \frac{\cos x}{\sin^2 x}\,dx &= \int \frac{1}{\sin x}\cdot\frac{\cos x}{\sin x}\,dx \\ &= \int \csc x \cot x\,dx = -\csc x + C\end{aligned}$$

2. $\displaystyle\int \frac{x^2}{x^2+1}\,dx$

$$\begin{aligned}\int \frac{x^2}{x^2+1}\,dx = \int \frac{x^2+1-1}{x^2+1}\,dx &= \int \left(\frac{x^2+1}{x^2+1} - \frac{1}{x^2+1}\right)dx \\ &= \int \left(1 - \frac{1}{1+x^2}\right)dx \\ &= x - \tan^{-1} x + C\end{aligned}$$

Exercise: Try Yourself!!!

Evaluate the following integrals:

1. $\displaystyle\int \frac{x^5 + 2x^2 - 1}{x^4}\,dx$
2. $\displaystyle\int (3\sin x - 2\sec^2 x)\,dx$
3. $\displaystyle\int \frac{\sin x}{\cos^2 x}\,dx$
4. $\displaystyle\int (1 + \sin^2\theta\,\csc\theta)\,d\theta$
5. $\displaystyle\int \left[\frac{1}{2\sqrt{1-x^2}} - \frac{3}{1+x^2}\right]dx$
6. *(Not now)* $\displaystyle\int \sqrt{16 - x^2}\,dx$
7. If $f'(x) = x^4 + 3x - 9$, what was $f(x)$?
8. $\displaystyle\int \sin\left(\frac{t}{2}\right)\cos\left(\frac{t}{2}\right)dt$

Integration by Substitution

Sometimes we cannot integrate straightforwardly. Then we substitute to another variable; this is called the **u-substitution** method. When we have a composite function and need to integrate it, we use the substitution method.

Suppose $F(g(x))$ is a composite function. Then

$$\frac{d}{dx}\big[F(g(x))\big] = F'(g(x))\,g'(x), \qquad \int F'(g(x))\,g'(x)\,dx = F(g(x)) + C.$$

Since F is an antiderivative of f, we write

$$\int f(g(x))\,g'(x)\,dx = F(g(x)) + C.$$

Substitution rule

Here we say

$$u = g(x), \qquad \frac{du}{dx} = g'(x), \qquad du = g'(x)\,dx,$$

which implies

$$\int f(u)\,du = F(u) + C.$$

Problem 1

Evaluate $\displaystyle\int 3(8y-1)\,e^{4y^2-y}\,dy$

Say $u = 4y^2 - y$, so $\dfrac{du}{dy} = 8y - 1$ and $du = (8y-1)\,dy$. Now,

$$\begin{aligned}\int 3e^{4y^2-y}(8y-1)\,dy = \int 3e^u\,du = 3\int e^u\,du = 3e^u + C \\ = 3e^{4y^2-y} + C\end{aligned}$$

Problem 2

Evaluate $\displaystyle\int x^2(3-10x^3)^4\,dx$

Say $u = 3 - 10x^3$, so $\dfrac{du}{dx} = -30x^2$, $du = -30x^2\,dx$, and $-\dfrac{1}{30}du = x^2\,dx$.

$$\begin{aligned}\int x^2(3-10x^3)^4\,dx = \int u^4\left(-\frac{1}{30}\right)du = -\frac{1}{30}\int u^4\,du \\ = -\frac{1}{30}\cdot\frac{u^5}{5} + C \\ = -\frac{1}{150}(3-10x^3)^5 + C\end{aligned}$$

Problem 3

Evaluate $\int \cos 5x \, dx$

Why substitute $u = 5x$*?* Because we integrate the integrand with respect to x, NOT $5x$.

Say $u = 5x$, so $\dfrac{du}{dx} = 5$ and $du = 5\,dx \Rightarrow \dfrac{1}{5}du = dx$.

$$\begin{aligned}\int \cos 5x \, dx &= \int \cos u \cdot \frac{1}{5}\, du = \frac{1}{5}\int \cos u \, du \\ &= \frac{1}{5}\sin u + C = \frac{1}{5}\sin 5x + C\end{aligned}$$

Problem 4

Evaluate $\int \sin^2 x \cos x \, dx$

Let $u = \sin x$, so $\dfrac{du}{dx} = \cos x$ and $du = \cos x \, dx$.

$$\int \sin^2 x \cos x \, dx = \int u^2 \, du = \frac{u^3}{3} + C = \frac{1}{3}\sin^3 x + C$$

Exercise: Evaluate the following integrals

1. $\displaystyle\int t^4\sqrt{3 - 5t^5}\, dt$
2. $\displaystyle\int \frac{e^x}{\sqrt{1 - e^{2x}}}\, dx$
3. $\displaystyle\int t\sqrt{7t^2 + 12}\, dt$
4. $\displaystyle\int \frac{x^2 + 1}{\sqrt{x^3 + 3x}}\, dx$
5. $\displaystyle\int e^{\sin x}\cos x \, dx$
6. $\displaystyle\int \frac{e^x + e^{-x}}{e^x - e^{-x}}\, dx$
7. $\displaystyle\int \frac{\sec^2 x}{\sqrt{1 - \tan^2 x}}\, dx$
8. $\displaystyle\int \frac{3^x}{(3^x + 1)^2}\, dx$